

Catastrophic Cancellation

Numerical Methods

David Mayerich

Scalable Tissue Imaging and Modeling (STIM) Laboratory

Department of Electrical and Computer Engineering

Cullen College of Engineering

University of Houston

UNIVERSITY of **HOUSTON** | ECE

Loss of Precision

Losing Digits to Cancellation

Loss of Precision Theorem

Catastrophic Cancellation

- Consider a floating point representation with a 5-digit mantissa:

$$\boxed{n} . \boxed{n} \boxed{n} \boxed{n} \boxed{n} \times 10^c$$

- Apply a simple addition and subtraction:

$$3.2342 \times 10^3 + 2.51 \times 10^{-1} - 3.2343 \times 10^3$$

$\begin{array}{r} 3234.2 \\ + 0.251 \\ \hline 3234.451 \end{array}$	$\longrightarrow$	$\begin{array}{r} 3234.451 \\ - 3234.3 \\ \hline 0.151 \end{array}$	$\begin{array}{r} 3234.5 \\ - 3234.3 \\ \hline 0.2 \end{array}$
round $\nearrow$ chop $\uparrow$			

Catastrophic Cancellation

- This subtraction effectively reduced our numerical precision:

$$x = 0.151 \quad \rightarrow \quad x = 0.2$$

creating a relative error of $\epsilon_r = \frac{|0.2 - 0.151|}{|0.151|} \approx 32.5\%$

- Each operation maintains the appropriate precision
- However, we ended up with far less precision than we expect:
 - The result $x = 0.2$ has one digit of precision
 - The correct value $x = 0.151$ requires two digits of precision
 - We have 5 digits of precision, expect 3, but only get 2
- This is known as *catastrophic cancellation*

Loss of Precision Theorem

Let x and y be normalized floating-point numbers such that $x > y > 0$. For a base b value, we can find two adjacent natural numbers $p, q \in \mathbb{N}$ such that:

$$b^{-p} \leq 1 - \frac{y}{x} \leq b^{-q} \quad \text{where} \quad q + 1 = p$$

At most p and at least q significant digits are lost in the subtraction $x - y$.

- Calculate the number of digits lost when calculating $x - y$ for:

$$x = 37.593621 \quad \text{and} \quad y = 37.584216$$

Lost Digits

$$b^{-p} \leq 1 - \frac{y}{x} \leq b^{-q}$$

$$\log(b^{-p}) \leq \log\left(1 - \frac{y}{x}\right) \leq \log(b^{-q})$$

$$-p \log(b) \leq \log\left(1 - \frac{y}{x}\right) \leq -q \log(b)$$

$$p \log(b) \geq -\log\left(1 - \frac{y}{x}\right) \geq q \log(b)$$

$$p \leq \frac{-\log\left(1 - \frac{y}{x}\right)}{\log(b)} \leq q$$

You will lose between $\left\lfloor \frac{-\log\left(1 - \frac{y}{x}\right)}{\log b} \right\rfloor$ and $\left\lceil \frac{-\log\left(1 - \frac{y}{x}\right)}{\log b} \right\rceil$

Loss of Precision Theorem

Calculate the logarithm of all terms

Separate exponents

Multiply by -1 (and reverse inequalities)

Divide by $\log(b)$

Lost Digits

- Calculate the number of digits lost when calculating $x - y$ for:

$$x = 37.593621 \quad \text{and} \quad y = 37.584216$$

$$p \leq \frac{-\log\left(1 - \frac{y}{x}\right)}{\log(10)} \leq q$$

$$p \leq \frac{-\log\left(1 - \frac{37.584216}{37.593621}\right)}{\log(10)} \leq q$$

$$p \leq \frac{8.2933482}{\log(10)} \leq q$$

$$p \leq 3.6017554 \leq q$$

$$p = \lfloor 3.6017554 \rfloor = 3 \quad q = \lceil 3.6017554 \rceil = 4$$

$$\begin{array}{r} 37.593621 \\ - 37.584216 \\ \hline 00.009405 \end{array}$$

Lost Bits

- Calculate the number of **bits** lost when calculating $x - y$ for:

$$x = 37.593621 \quad \text{and} \quad y = 37.584216$$

$$p \leq \frac{-\log\left(1 - \frac{y}{x}\right)}{\text{ }} \leq q$$

$$p \leq \frac{-\log\left(1 - \frac{37.584216}{37.593621}\right)}{\text{ }} \leq q$$

$$p \leq \frac{8.2933482}{\text{ }} \leq q$$

$$p \leq 11.964772 \leq q$$

$$p = \lfloor 11.964772 \rfloor = 11 \quad q = \lceil 11.964772 \rceil = 12$$

$$x = 100101.10010111111101111001$$

$$y = 100101.10010101100011110011$$

$$\text{ } 10011010000110$$

Discussion: Precision After Subtraction

- How many bits of precision are lost when calculating $\alpha - \sin \alpha$ if $\alpha = \frac{1}{64}$?

$$p \leq \frac{-\log \left(1 - \frac{\sin\left(\frac{1}{64}\right)}{\frac{1}{64}} \right)}{\log(2)} \leq q$$

$$p \leq 14.58 \leq q$$

$$p = \lfloor 14.58 \rfloor = 14$$

$$q = \lceil 14.58 \rceil = 15$$

- What precision can we expect if this calculation is performed on `float32` registers?

IEEE 754 Standard

standard	C/C++	m bits	x bits	bias
binary16	single	10	5	15
binary64	double	53	11	1023
binary128	N/A	113	15	16383
binary256	N/A	19	237	262143

$$23 - 15 = 8$$

Recovering Precision

Reformulation

Verifying Reformulations

The Quadratic Equation

Summary of Arithmetic Operations

- Given d digits of precision:
 - multiplication:** preserves d digits because the exponent and mantissa can be calculated independently
 - addition:** exponents must be equal, so significant digits may be *chopped*
- Evaluate $x = 1 - \sin 1.6$ using three digits of precision:

expected result (with 3 digits)

$$y = 1.00 - \sin 1.61$$

$$y = 1.00 - 0.9992316$$

$$y = 1.00 - 0.999 = 0.00$$

$$y = 1 - 0.9992316 = 0.0007684$$

$$\epsilon_a = |0.00100 - .000768| = .000232$$

$$\epsilon_r = \frac{|0.000232|}{|0.000768|} = 0.30 \quad (30\%)$$

Reformulation

- Precision loss appears after subtraction (but the loss of data occurs earlier)
- Reformulate expressions to remove subtractions

$$y = 1 - \sin x$$

$$y = (1 - \sin x) \left(\frac{1 + \sin x}{1 + \sin x} \right)$$

$$y = \frac{1 - \sin^2 x}{1 + \sin x} \quad \swarrow \quad \sin^2 x + \cos^2 x = 1$$

$$y = \frac{\cos^2 x}{1 + \sin x}$$

$$y = \frac{\cos^2 1.61}{1 + \sin 1.61}$$

$$y = \frac{(-0.0391936)^2}{1.00 + 0.9992316} \longrightarrow y = \frac{(-0.0392)^2}{1.00 + 0.999}$$

$$y = \frac{0.00153664}{1.999} \longrightarrow y = \frac{0.00154}{2.00}$$

$$y = .000$$

$$\epsilon_r = \frac{|.000768 - .00077|}{|.000768|} = 0.0026 \quad (0.26\%)$$

Reformulation (Part Deux)

- When can catastrophic cancellation occur in the expression $\sqrt{x+1} - \sqrt{x}$?
- When x is large -- fix it for $x = 10$:

$$y = (\sqrt{x+1} - \sqrt{x}) \left(\frac{\sqrt{x+1} + \sqrt{x}}{\sqrt{x+1} + \sqrt{x}} \right)$$

Expected Result:

$$y = 0.154347 = 0.154$$

$$y = \left(\frac{x+1-x}{\sqrt{x+1} + \sqrt{x}} \right)$$

$$y = \frac{1}{\sqrt{x+1} + \sqrt{x}}$$

Original Expression:

$$y = \sqrt{11} - \sqrt{10}$$

$$y = 3.31662 - 3.16228$$

$$y = 3.32 - 3.16 = 0.160$$

$$\epsilon_r = \frac{|0.154 - 0.160|}{|0.154|} \approx 3.9\%$$

Reformulation:

$$y = \frac{1.00}{3.32 + 3.16}$$

$$y = \frac{1.00}{6.48} = 1.54321$$

$$\epsilon_r = 0\%$$

Quadratic Equation

Loss of Precision

Reformulation

Implementation

Quadratic Equation

- Find the roots of a polynomial $ax^2 + bx + c = 0$
- Quadratic Equation:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- Two cases cause computational problems:
 1. If $a = 0$, then the equation is undefined (the polynomial isn't quadratic)
 2. If $4ac \ll b^2$ then catastrophic cancellation can occur in the numerator

Two cases for cancellation:

- if $b > 0$ then $-b + \sqrt{b^2 - 4ac}$ can result in cancellation
- if $b < 0$, then $b - \sqrt{b^2 - 4ac}$ can result in cancellation

Testing the Quadratic Equation

- Consider the roots of the polynomial $x^2 + 182^3x - 1$ (using three significant digits):

$$x_1 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

$$x_2 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$x_1 = \frac{-182^3 - \sqrt{182^6 - 4}}{2}$$

$$x_2 = \frac{-182^3 + \sqrt{182^6 - 4}}{2}$$

- Performing this calculation using IEEE 754 floating point, we get:

binary64

$$x_1 = -6.0286 \times 10^8$$

$$x_1 = -1.657865 \times 10^{-7}$$

binary32

$$x_1 = -6.0286 \times 10^8$$

$$x_1 = -1.5 \times 10^{-7}$$

$$\epsilon_r \approx 9.5\%$$

Reformulating the Quadratic Equation

- When $b \gg 4ac$ there is one *very* small root
- Assume that $b > 0$ and reformulate the "small" root:

$$x_2 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$x_2 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \cdot \frac{-b - \sqrt{b^2 - 4ac}}{-b - \sqrt{b^2 - 4ac}}$$

$$x_2 = \frac{b^2 - (b^2 - 4ac)}{2a(-b - \sqrt{b^2 - 4ac})}$$

$$x_2 = \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$

Simplifying the Reformulation

- For the case where $b > 0$ we have two roots:

$$x_1 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

(large root)

$$x_2 = \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$

(small root)

$$-2ax_1 = b + \sqrt{b^2 - 4ac} \longrightarrow x_2 = \frac{-2c}{-2ax_1}$$

$$x_1 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

$$x_2 = \frac{c}{ax_1}$$

Implementing the Quadratic Equation

- Consider the reformulation of both cases for b :

$$b > 0 \quad x_1 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

$$x_2 = \frac{c}{ax_1}$$

$$b \leq 0 \quad x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$x_2 = \frac{c}{ax_1}$$

$$q = -\frac{1}{2} \left[b + \operatorname{sgn}(b) \sqrt{b^2 - 4ac} \right] \quad \text{where} \quad \operatorname{sgn}(x) = \begin{cases} -1 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$

$$x_1 = \frac{q}{a}$$

$$x_2 = \frac{c}{q}$$